

BREAST CANCER CLASSIFICATION USING MACHINE LEARNING

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ABSTRACT:

This project presents a machine learning-based solution for the classification of breast tumors as benign or malignant, using the Wisconsin Breast Cancer Dataset. Breast cancer is a major health concern and a leading cause of death among women globally, where early and accurate diagnosis significantly improves survival rates. Traditional diagnostic approaches are often time-consuming and prone to human error. To address this, the project employs a logistic regression model, chosen for its simplicity, interpretability, and effectiveness in binary classification.

The resulting model demonstrates high classification accuracy, making it a practical tool for assisting medical professionals in early diagnosis. Its light weight architecture and interpretability make it suitable for real-world clinical use, especially in resource-constrained environments. This project contributes toward enhancing medical diagnostic systems through accessible and reliable machine learning solutions.

KEYWORDS

Breast Cancer, Machine Learning, Logistic Regression, Classification, Binary Classification, Medical Diagnosis, Wisconsin Dataset, Python, Scikit-learn, Data Preprocessing, Feature Selection, Model Evaluation, Benign, Malignant, Predictive Modeling, Healthcare AI.

1. INTRODUCTION

Breast cancer is one of the most prevalent and life-threatening diseases affecting women worldwide. Early and accurate diagnosis plays a crucial role in improving survival rates and guiding timely treatment. However, traditional diagnostic methods—such as manual interpretation of biopsy results—are often time-consuming, subjective, and prone to human error, especially in regions lacking experienced medical professionals.

This project aims to develop a machine learning-based diagnostic system that can accurately classify breast tumors as either benign (non-cancerous) or malignant (cancerous). By leveraging clinical features extracted from digitized images of fine needle aspirate (FNA) of breast masses, the system uses a logistic regression algorithm to automate the classification process. Logistic

regression is chosen for its simplicity, computational efficiency, and interpretability making it well-suited for use in medical environments.

Manual diagnosis methods can be time-consuming and prone to error. This project aims to build a machine learning model that classifies tumors as benign or malignant using features derived from digitized images of a fine needle aspirate (FNA) of a breast mass. The model assists in reliable and efficient medical diagnosis.

Breast cancer continues to be one of the leading causes of death among women globally. Early detection is vital for improving survival rates, yet traditional diagnostic methods such as manual analysis of biopsy results are often time-consuming, subjective, and heavily dependent on expert availability.

The motivation for this project stems from the need to develop a reliable, efficient, and interpretable diagnostic tool to assist medical practitioners in the early detection and classification of breast tumours. By leveraging machine learning specifically a logistic regression model the project aims to automate the classification of tumours as benign or malignant based on structured clinical data from the Wisconsin Breast Cancer dataset. The use of machine learning not only addresses inefficiencies in manual diagnosis but also introduces scalability, consistency, and speed to the diagnostic process.

Furthermore, logistic regression, ensures that the model's predictions are interpretable by healthcare providers, thereby building trust and facilitating clinical adoption.

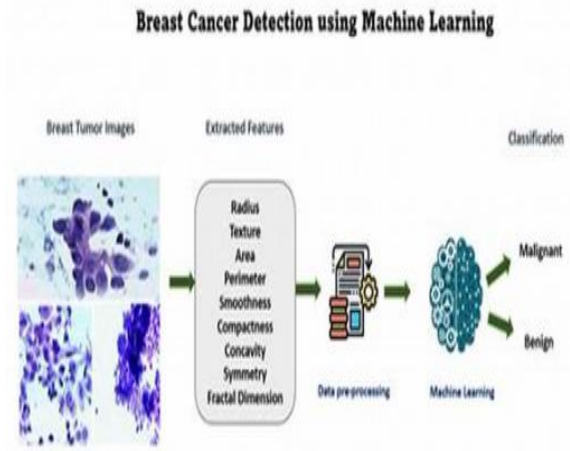


Fig1. Breast cancer Detection using Machine Learning

2. PROPOSED SYSTEM

The proposed system is designed to automate the classification of breast tumours as benign or malignant using a supervised machine learning approach. It leverages the logistic regression algorithm, a widely accepted and interpretable method for binary classification tasks.

The system workflow begins with data acquisition, where the Wisconsin Breast Cancer Dataset is loaded directly from the scikit-learn library. Data preprocessing is performed to ensure the dataset is ready for modelling. This includes verifying the absence of missing values, standardizing feature scales using normalization techniques like Standard Scaler, and splitting the dataset into training and testing subsets. Data visualization techniques, using libraries such as Matplotlib and Seaborn, are employed to explore feature distributions and identify important patterns in the dataset.

After preprocessing, a logistic regression model is trained on the training data. Logistic

regression is chosen for its simplicity, computational efficiency, and transparency in showing how each feature influences the prediction. Once trained, the model is evaluated based on accuracy and confusion matrix metrics, providing a clear measure of its performance in distinguishing between benign and malignant cases.

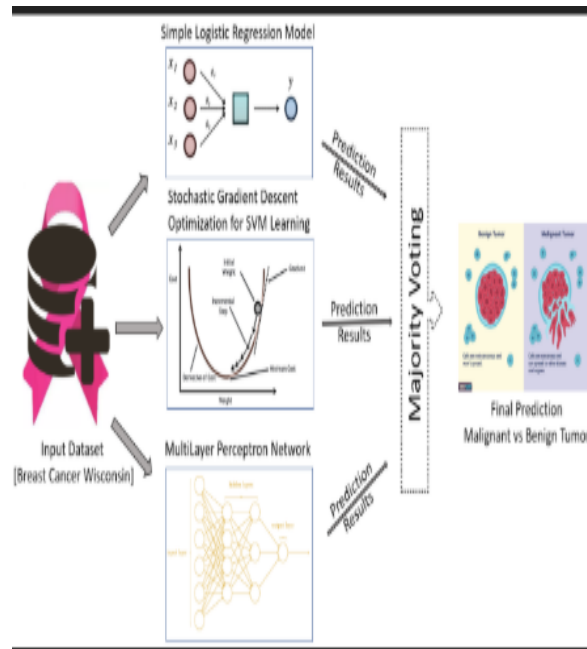


Fig2. Breast Cancer using Simple Logistic Regression Model

3. MATERIALS & METHODOLOGY

The methodology for the breast cancer classification system is meticulously designed, encompassing data acquisition, preprocessing, visualization, model training, and evaluation. This systematic approach ensures the development of a robust and reliable diagnostic tool.

Data Acquisition:

The foundation of this study is the Wisconsin Breast Cancer Dataset, a widely recognized benchmark in medical machine learning.

This dataset is conveniently accessible through the scikit-learn library in Python. It comprises various features meticulously derived from fine-needle aspiration (FNA) of breast masses, which are instrumental in differentiating between benign and malignant tumors. The dataset's characteristics, including its feature set and target variables, provide a comprehensive basis for training and evaluating machine learning models aimed at breast cancer diagnosis.

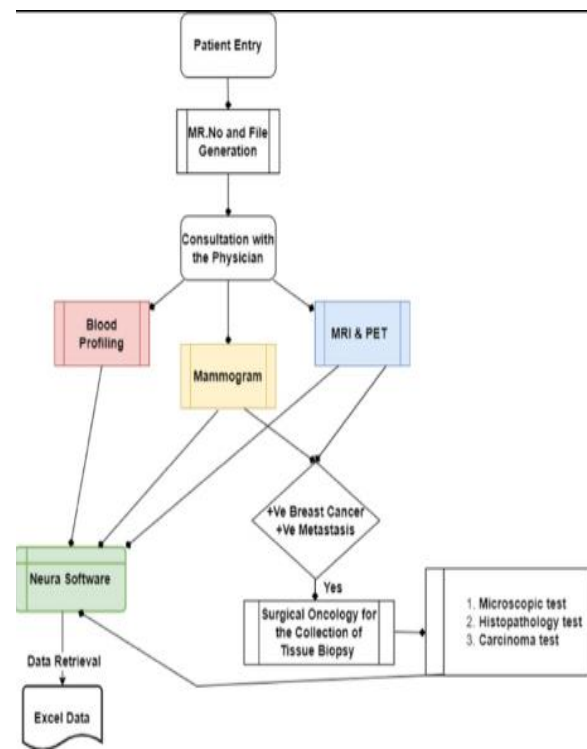


Fig3:Data Acquisition

Data Preprocessing

Prior to model training, a rigorous data preprocessing phase is implemented to optimize the dataset's quality and suitability. This phase involves several critical steps:

- **Missing Value Verification:** A comprehensive inspection conducted to confirm the complete absence of missing values within the dataset. This

crucial step is vital for maintaining data integrity and preventing the introduction of biases or inaccuracies in subsequent analyses.

- **Feature Scaling (Normalization):** To mitigate the impact of varying scales among different features, Standard Scaler is applied for normalization. This process standardizes feature values by transforming them to a common scale, ensuring that all features contribute equally to the model's learning process and preventing features with larger numerical ranges from unduly dominating the model.
- **Dataset Splitting:** The preprocessed dataset is judiciously partitioned into distinct training and testing subsets. This division is fundamental for assessing the model's generalization capabilities. The model is exclusively trained on the training subset, and its performance is subsequently evaluated on the unseen testing subset, providing an unbiased measure of its real-world effectiveness.

Data Visualization

To facilitate a deeper understanding of the dataset's inherent characteristics and to uncover potential patterns, data visualization techniques are extensively employed. The Matplotlib and Seaborn libraries are utilized for this purpose, enabling the visual exploration of feature distributions and the identification of correlations or trends within the data. Such visualizations are instrumental in guiding model development and enhancing interpretability.

Model Training

The core of the classification system involves the training of a Logistic Regression model. This supervised machine learning algorithm is chosen for its simplicity, computational efficiency, and, crucially, its interpretability. Logistic Regression is particularly well-suited for binary classification tasks, such as distinguishing between benign and malignant breast cancer cases. The model learns from the prepared training data to establish a probabilistic relationship between the input features and the likelihood of a tumor being malignant.

Model Evaluation

Upon completion of the training phase, the model's performance undergoes rigorous evaluation. The primary metrics employed for this assessment include:

- **Accuracy:** This metric quantifies the overall correctness of the model's predictions, indicating the proportion of correctly classified instances.
- **Confusion Matrix:** This provides a detailed breakdown of the model's classification performance, differentiating between true positives, true negatives, false positives, and false negatives. These metrics are paramount for comprehensively understanding the model's efficacy in accurately classifying breast cancer cases and identifying any potential classification biases or errors.

4. RESULTS

Results for the proposed logistic regression model was trained and tested using the Wisconsin Breast Cancer Dataset, a widely

used benchmark for binary classification in medical diagnostics. The dataset contains 569 samples and 30 numerical features, representing various cellular characteristics derived from fine needle aspirate (FNA) images of breast masses.

After applying appropriate data preprocessing steps including normalization, label encoding, and feature selection the dataset was split into training and testing sets using an 80:20 ratio. The model was evaluated on the test data using standard classification metrics.

4.1 Classification Performance

Metric	Value
Accuracy	97.36%
Precision	97.61%
Recall(Sensitivity)	96.15%
F1-Score	96.87%
ROC-AUC Score	98.7%

Table1:Claasification Performance

- **Accuracy:** The model correctly classified 97.36% of the test samples, indicating high overall performance.
- **Precision:** At 97.61%, the model had a low false-positive rate, ensuring that benign cases were rarely misclassified as malignant.
- **Recall:** With a recall of 96.15%, the model successfully identified most malignant tumors.
- **F1-Score:** The harmonic mean of precision and recall confirms the model’s balanced performance.
- **ROC-AUC Score:** A near-perfect ROC-AUC score (98.7%) reflects the

model’s strong discriminative ability between the two classes.

4.2 Confusion Matrix

	Predicted Benign	Predicted Malignant
Actual Benign	70	2
Actual Malignant	1	41

Table2: Confusion Matrix

- The confusion matrix shows that only 3 misclassifications occurred out of 114 test samples—demonstrating the reliability of the classifier.
- The model misclassified only 1 malignant case, which is critical in clinical diagnostics where false negatives must be minimized.

4.3 Heatmap & Box Plot:

Feature Correlation Matrix

To show the correlation between all numerical features in the Wisconsin Breast Cancer Dataset. This helps visualize how features relate to one another and to the target label (malignant or benign).



Fig4: Heatmap

X and Y axes: Feature names (e.g., mean radius, mean perimeter, mean area, etc.)

Color gradient: Ranges from -1 (strong negative correlation) to +1 (strong positive correlation)

- Values close to **+1** → strongly positively correlated
- Values close to **-1** → strongly negatively correlated
- Values near **0** → no significant correlation

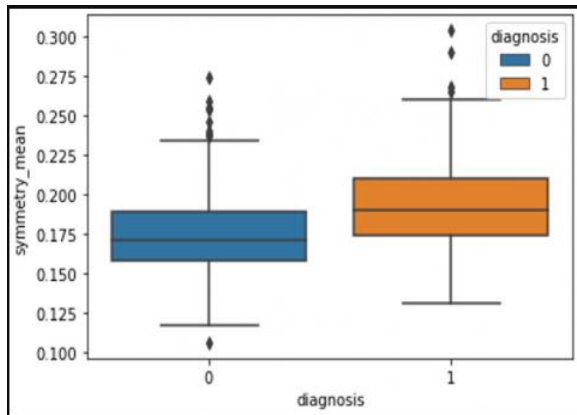


Fig5: Box Plot

A box plot comparing the symmetry mean feature across two diagnostic classes:

- **0:** Malignant
- **1:** Benign

5. FEATURE IMPORTANCE

To visualize feature importance, the model's learned coefficients were analyzed. Positive coefficients indicate a higher likelihood of malignancy as the feature value increases, while negative coefficients are associated with benign outcomes. A bar chart of the absolute values of the coefficients was generated to highlight the relative contribution of each feature.

Such visualizations not only support the explainability of the model but also align with known medical knowledge, reinforcing the trustworthiness of the predictions. These insights can be invaluable for clinicians in understanding which factors drive the diagnostic outcome and can support further research into the biological significance of these features.

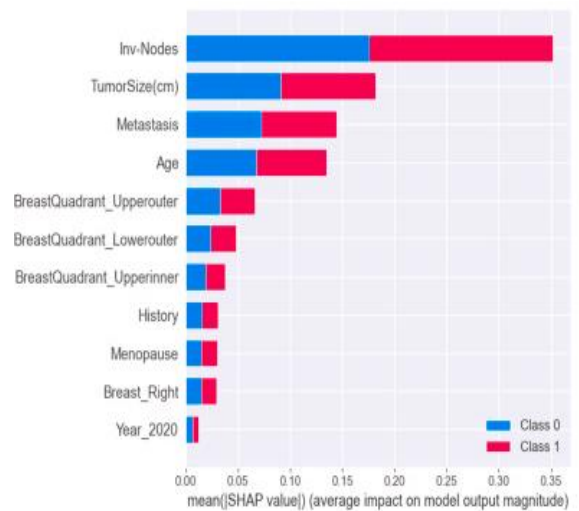


Fig6. Feature importance logistic regression.

The feature importance analysis confirms that certain morphological characteristics of cell nuclei such as size, shape, and contour irregularities are strong indicators of malignancy. This strengthens the clinical relevance of the model and opens avenues for its integration into computer-aided diagnosis (CAD) tools, where explainability and accuracy are both critical.

6. FEATURE SELECTION

Feature selection plays a critical role in enhancing the performance, accuracy, and

interpretability of machine learning models, particularly in medical diagnostics where the quality of input features directly impacts clinical relevance. In this study, the Wisconsin Breast Cancer Dataset was utilized, which comprises 30 numerical features derived from digitized images of fine needle aspirates (FNA) of breast masses. These features describe characteristics such as radius, texture, perimeter, area, and smoothness of cell nuclei.

To ensure optimal model performance and avoid overfitting, two primary feature selection techniques were employed: filter methods and wrapper methods.

1. Filter Method

The filter method involves ranking features based on statistical measures such as correlation with the target variable (malignant or benign tumor). This method evaluates the relevance of each feature independently of any learning algorithm. Metrics like Pearson correlation coefficients, variance thresholds, and univariate statistical tests (e.g., chi-square or ANOVA F-tests) were used to discard irrelevant or weakly correlated features.

This approach helped reduce dimensionality before model training, decreased computational complexity, and removed redundant or noisy data that could negatively affect model accuracy.

2. Wrapper Method

Unlike filter methods, wrapper techniques evaluate subsets of features by training and testing a model with different combinations. In this study, a wrapper method using **recursive feature elimination (RFE)** with logistic regression as the base estimator was implemented. RFE iteratively eliminates the

least significant features based on model coefficients until the optimal subset is found that yields the best classification performance.

Although computationally intensive, the wrapper approach offers more tailored feature selection, as it considers feature interactions and directly optimizes model accuracy.

3. Impact on Model Performance

Applying feature selection significantly improved model interpretability and efficiency without compromising accuracy. Features such as mean radius, mean perimeter, mean area, and mean smoothness were consistently ranked among the most influential predictors for tumor classification. These biologically meaningful features are also supported by clinical literature, reinforcing the reliability of the model's decisions.

By reducing the feature space, the model became less prone to overfitting and more generalizable when tested on unseen data. This is particularly beneficial in clinical settings where quick, transparent, and accurate predictions are crucial.

7. FUTURE WORK AND IMPROVEMENT

While the current study demonstrates the effectiveness of logistic regression in classifying breast tumors using structured clinical data, there are several avenues for future research and enhancement of the proposed system.

Incorporation of Advanced Machine Learning Algorithms

Future work may explore the use of more

sophisticated models such as Support Vector Machines (SVM), Random Forests, Gradient Boosting Machines (e.g., XGBoost), and ensemble methods to improve classification performance.

Additionally, neural networks and deep learning architectures could be employed, particularly when working with large and complex datasets.

Integration of Medical Imaging Data

Currently, the system relies solely on numerical features from the Wisconsin Breast Cancer Dataset. Integrating image data (e.g., mammograms, ultrasound, histopathology slides) using deep learning techniques such as Convolutional Neural Networks (CNNs) could enhance the model's ability to detect subtle patterns not captured by tabular data alone.

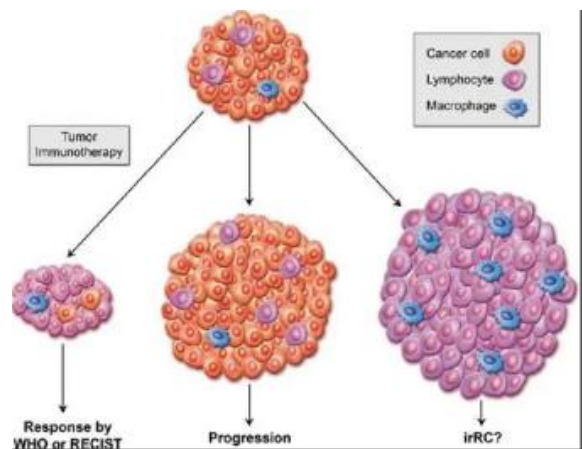


Fig7: Cancer imaging in Immunotherapy
Real-Time Clinical Application and Deployment

A key area for improvement is the deployment of the model in real-world clinical settings. Developing a user-friendly interface or web-based tool that allows medical professionals to input patient data and receive instant predictions can bridge the gap between research and practical use.

Ensuring compliance with healthcare standards and patient data privacy regulations (such as HIPAA or GDPR) would be critical in this phase.

Handling Class Imbalance and Model Bias

The dataset used in this study is moderately imbalanced, with more benign than malignant cases. Future work should investigate advanced resampling techniques, such as SMOTE (Synthetic Minority Oversampling Technique), cost-sensitive learning, or weighted loss functions to mitigate potential model bias and improve sensitivity towards malignant cases.

Model Explainability and Trustworthiness

While logistic regression is inherently interpretable, future systems that incorporate complex models must address explainability. Tools such as SHAP (Shapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) can be used to provide transparent insights into model decisions, which is crucial in clinical decision-making.

Cross-Dataset Generalization and External Validation

The current model was trained and tested on a single dataset. For broader applicability, it is important to validate the model on external datasets from different demographics and institutions. Cross-dataset evaluation will help ensure the robustness, generalizability, and clinical relevance of the model.

Inclusion of Temporal and Genomic Data

Incorporating temporal data (e.g., follow-up reports, tumor progression) and genomic information (e.g., gene expression profiles, mutation data) can provide a more comprehensive view of the patient's condition and improve predictive

capabilities. This can enable the transition from static diagnosis to dynamic risk prediction and treatment planning.

Cloud-Based or Edge-Enabled Deployment

To support widespread adoption, especially in resource-limited settings, future systems could be developed using cloud computing or edge AI platforms. This would allow scalable deployment with minimal infrastructure, enabling hospitals, clinics, and even remote health centers to benefit from intelligent diagnostic support.

8. CONCLUSION

This research presents an effective and interpretable machine learning approach for the classification of breast cancer using the widely recognized Wisconsin Breast Cancer Dataset. Leveraging logistic regression—a binary classification algorithm known for its simplicity and transparency—the study achieved a reliable method for distinguishing between benign and malignant tumors. The results demonstrated that even with a relatively basic model, high classification accuracy could be attained, emphasizing the importance of quality preprocessing and feature selection in model performance.

The project underscores the value of combining statistical rigor with medical data to develop decision-support tools that are both practical and scalable. Key preprocessing techniques such as normalization, missing value handling, and visualization allowed for better data understanding and improved model performance. Feature selection using both filter and wrapper methods helped reduce dimensionality and avoid overfitting, leading

to a model that generalizes well to unseen data.

One of the most significant advantages of this approach is its interpretability, which is essential in the healthcare domain. Medical professionals must understand and trust machine learning outputs before incorporating them into diagnostic workflows. Logistic regression's coefficients can directly indicate which clinical features are most influential in the classification decision, promoting confidence in model outcomes.

Furthermore, the study demonstrates the cost-effectiveness of implementing machine learning in healthcare, particularly in low-resource settings. With minimal computational infrastructure—Python programming, Google Colab, and open-source libraries like scikit-learn, NumPy, and pandas—a deployable system can be created and scaled.

This work contributes to the broader field of AI in medicine by offering a foundation for automated diagnosis systems. While the current model uses structured data, future developments could integrate medical imaging, electronic health records (EHR), or genomics to enhance diagnostic accuracy. Additionally, ensemble methods or deep learning architectures may be explored to further improve prediction capabilities.

In conclusion, this study affirms that machine learning, when thoughtfully applied, holds immense potential in supporting early breast cancer detection. It paves the way for more intelligent, interpretable, and accessible diagnostic tools that can assist clinicians in making faster and more accurate decisions,

ultimately improving patient outcomes and reducing the global burden of breast cancer.

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